** Scale :-

→ Scale is the technique through which ground distances are represented on map/paper.

$$\boxed{\text{Scale} = \frac{\text{Map distance}}{\text{Ground distance}}}$$

Scale

- Numerical Scale
  - Engineer's Scale
    (1 cm = ground distance)
    (1 cm = 10 km)
  - Representative fraction (RF)
    (It is fractional from the scale)
    Eg:- $R.F = \frac{1\ cm}{10 \times 10^3 \times 100\ cm} = \frac{1}{10^6}$
- Graphical Scale

Q) The Engineer's scale of a drawing, is stated to be 1 cm = 4m, Convert this to fraction scale.

Soln

Scale; 1 cm = 4m

$$R.F = \frac{1}{4 \times 100} = \frac{1}{400}$$

Q) The fraction scale of a map is stated to the 1:50000 Convert this to engineer's scale.

Soln:

$$R.F = \frac{1}{50000}$$

mean (1 cm on plan = 50000 cm in ground)

1 cm on plan = 500 m on ground

∴ Engineer's Scale

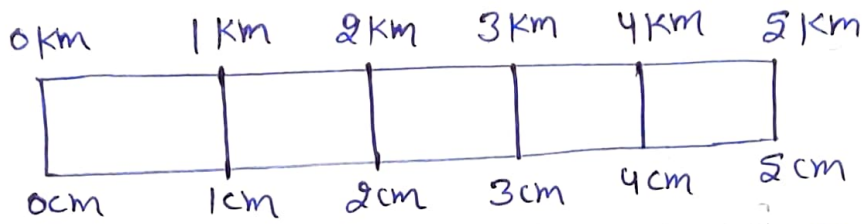
$$\boxed{1\,cm = 500\,m}$$

* Graphical Scale:-

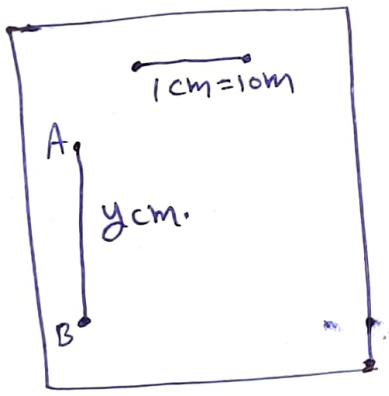
→ Graphical scale is a line drawn on the plan or map on which ground distance is directly marked.

→ Graphical scale has the advantage over numerical scale such that distance on the plan or map can be determine by actual scale even through plan or map has shrunk.

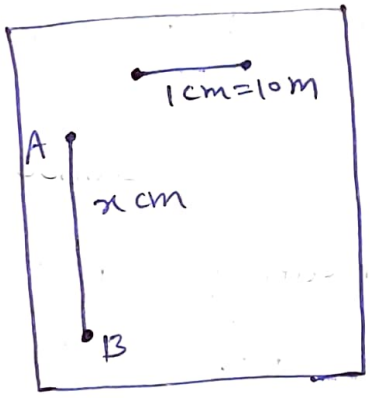
→ In case of shrinkage of map graphical scale also changes with the map and therefore ratio is unaffected.



* Errors incurring due to shrinkage of map:-



After Shrinkage



Before Shrinkage

Before Shrinkage

Scale = (1 cm = 10 m)

Also, let distance between two points on the map = x cm

$\Rightarrow$ x cm on the map represents 1000x cm on ground.

After Shrinkage:-

Now distance between the two points = y cm.

$\therefore$ ground distance = 1000 y cm

but

$$\text{Shrunk Scale} = \frac{y \text{ cm}}{1000x \text{ cm}} = \left(\frac{y}{x}\right)\cdot\left(\frac{1}{1000}\right)$$

$$\text{Shrunk Scale} = (\text{Shrinkage factor}) \times \underset{\substack{\uparrow \\ \text{original scale.}}}{\text{R.F}}$$

Where y = shrunk length
x = original length.

$$\boxed{\text{Shrinkage factor} = \frac{y}{x}} = \frac{\text{map distance after shrinkage}}{\text{map distance before shrinkage}}$$

$$\boxed{\text{Shrinkage factor} = \frac{\text{Shrunk length}}{\text{original length}} = \frac{\text{Shrunk Scale}}{\text{original Scale}} = \frac{\text{Shrunk R.F}}{\text{original R.F}}}$$

Note:- $\boxed{\text{Shrinkage factor always} < 1}$

Q) plan of an area has shrunk such that a line originally 10 cm long now measures as 9.5 cm. If original scale was 1 cm = 10 m, determine.

a). Correct length corresponding to measured length of 100 m.

b) Correct area corresponding to measured area of 100 m^2.

c) Shrinkage factor

d) Shrunk scale.

Soln:-

Scale 1 cm = 10 m.

$$\boxed{\text{Shrinkage factor, } S.F = \frac{9.5}{10} = 0.95}$$

$$\text{Shrunk Scale} = S.F \times \text{original Scale (R.F)}$$
$$= 0.95 \times \frac{1}{1000} = \frac{1}{1052.63}$$

$$\boxed{\text{Shrunk Scale} = \frac{1}{1052.63}}$$

$$S.F = \frac{\text{Shrunk length}}{\text{original length}}$$

$$\Rightarrow \text{original length} = \frac{\text{Shrunk length}}{S.F} = \frac{100}{0.95} = 105.263\,m$$

$$\Rightarrow \boxed{\text{original length} = 105.263\ m}$$

Now

$$(S.F)^2 = \frac{\text{measured Area}}{\text{Correct Area}}$$

$$\therefore \text{Correct Area} = \frac{\text{measured Area}}{(S.F)^2} = \frac{100}{(0.95)^2} = 110.803\,m^2$$

$$\Rightarrow \boxed{\text{Correct Area} = 110.803\,m^2}$$

—x—x—